Teaching Philosophy Statement

Adithya Sathyanarayana

My primary goal as an instructor is to make sure every student feels safe, welcome, and respected, and knows they belong in the mathematics classroom. Students arrive with very different experiences of mathematics, and some have had far more support and encouragement than others. I share the convictions behind Federico Ardila-Mantilla's axioms for mathematics education: that mathematical talent is distributed equally across groups of people, and that every student deserves meaningful mathematical experiences and to be treated with dignity and respect.[1]

My own interest in mathematics took time to develop. As a child, I struggled with mathematics and physics, but I still vividly remember the experiments shown to us by a physics teacher who made a two-hour journey through Indian city traffic to teach our weekend classes. Only later, as I solved more problems and began asking my own questions, did I fall in love with the subject. That teacher showed me how much an instructor can do by making room for curiosity and believing that students can understand. His example is why I volunteer with math circles and with science and mathematics camps for girls, and it underlies three commitments that run through my teaching: students should do mathematics, not just watch it; they should do it together; and they should be cared for as whole people.

I have carried these commitments across a wide range of courses and formats at Michigan State University (MSU). I have been instructor of record for College Algebra I (MTH 103A), Survey of Calculus (MTH 124), Matrix Algebra with Applications (MTH 314), and Multivariable Calculus (MTH 234), and a teaching assistant for courses from introductory proofs (MTH 299) and honors linear algebra (MTH 317H) to graduate algebra (MTH 818). I have taught in person, fully online, and in person with a live online option. This statement describes the principles behind that work; my teaching portfolio and website, sites.google.com/msu.edu/adithya, contain the course materials and fuller detail.

Doing mathematics, not just watching it. Because I came to love mathematics by working through problems myself, I build class time around students' own attempts. After introducing a concept and working through an example, I give students time to try a related problem before I present a solution, and I circulate, offering small hints and asking questions that help them find a next step. When a student is unsure or makes a mistake, my first concern is to help them feel at ease: I remind the class that an incomplete idea is still worth discussing, and I openly acknowledge my own mistakes. One student wrote, "He treats all the students respectfully and replies to our queries quickly," and said that the encouraging environment helped them learn. I also look for the right picture, since it often makes sense before a formal statement does. In MTH 314, I used geometric transformations of the plane so that students would understand what eigenvalues and eigenvectors tell us about a linear transformation while they learned to compute them. Another wrote that "we learned what these concepts actually represented in linear algebra rather than just the routine calculations."

Learning together, in person and online. Explaining an idea to a classmate is one of the best ways to understand it, and asking a classmate is often easier than asking the instructor. In Spring 2025, I taught about sixty students in MTH 103A, College Algebra I, in an active-learning classroom in the MSU Union with round tables and screens throughout the room. Students spent much of each class working in groups at their tables while two undergraduate learning assistants and I circulated to help, and I wrote live on an iPad, projected onto several screens so that every table could follow along. Online, collaboration is much harder: students rarely get the chance to meet their peers, let alone work with them. When I taught MTH 124, Survey of Calculus, online in Summer 2023, I structured it as a flipped classroom largely for this reason. Students watched videos I recorded as pre-class assignments, meeting new ideas at their own pace, so that our live sessions could be spent together: working through problems, discussing their solutions, and

[1] Federico Ardila-Mantilla, *Todos Cuentan: Cultivating Diversity in Combinatorics*, Notices of the AMS **63** (2016), no. 10, 1164–1170.

consolidating the main theory. The course also ran on Piazza, where students earned engagement points for participating and asked a great many questions. To help classmates get to know one another, I invited students to make a short "reel" introducing themselves. This was entirely optional, and students shared only what they were comfortable sharing. The reels helped students meet their peers and find people to work with, both in class and, if they wished, outside it.

Caring for the whole student. My students have lives, responsibilities, and pressures outside my course, and they learn best when that is acknowledged. When a student faces difficult circumstances, I offer flexible deadlines so they can complete the work rather than fall behind for reasons beyond their control, and I pay attention to how students are doing, not only to how they are performing. I plan the course calendar with their preparation in mind: I space exams so that students have time to prepare, and I avoid scheduling one right after a long weekend or break, which should be a time to rest. An exam is a fair measure of what a student has learned only when they have had a fair chance to prepare for it. The same concern shaped my use of technology. MTH 103A met every day at 8 a.m., and part of the semester fell in the depths of a Michigan winter. I did not want students to choose between staying safe on icy mornings and keeping up with the class, so I offered a Zoom option for those who could not make it to campus and for days when the university might close. In MTH 314, which ran fully online in Summer 2024, a struggling student can easily go unnoticed and feel alone, so I did not wait for students to come to me: I wrote to those who were not doing well and met with them individually on Zoom. Standards-based grading in MTH 103A served a similar purpose: students had further opportunities to demonstrate mastery of topics they had initially found difficult, so an unsuccessful attempt became a reason to keep working rather than a final verdict.

Designing materials for these goals. I write my own learning materials, and each is designed with a purpose. For MTH 314, I wrote lecture notes typeset in LaTeX and also shared the handwritten notes I produced live in each Zoom lecture, so students had both a polished reference and a record of how each argument unfolded. I recorded lecture videos, posted short check-in videos between meetings, and designed in-class Python assignments so students practiced the computational side of matrix algebra alongside the theory. For MTH 103A, I prepared my own lecture notes and worksheets. As a teaching assistant for MTH 317H, Honors Linear Algebra, I teach the proof techniques from MTH 299 that students need but may not yet know, with worksheets that ask why a concept is useful or needed, and I release complete solutions twenty-four hours after each deadline so students can check their reasoning. I also bring generative AI into proof-writing recitations, giving students poorly written AI-generated proofs and asking them to find unjustified assumptions, missing steps, and errors that are easy to overlook when an explanation sounds convincing, and to say what the AI did well. I make clear that AI should not do their homework; my aim is for students to understand its possibilities and limitations and to gain the confidence to assess an argument for themselves.

Continuing to grow as a teacher. I am completing MSU's Certification in College Teaching (expected Spring 2027), which includes the course MTHE 879, Teaching College Mathematics, and the Certification in College Teaching Institute, with workshops on learning environments, assessment, and technology in teaching. Conversations with classmates from several disciplines have encouraged me to give students more choice in how they show understanding. Where a course allows it, I would like to offer projects such as a video, artwork, or a physical model, like a toy that demonstrates projectile motion for a unit on quadratics. I also learn from watching others teach. As a lead teaching assistant in 2024–25, I observed and gave feedback to eight teaching assistants and sixteen undergraduate learning assistants, and I have spent six semesters as a tutor and supervisor in the Mathematics Learning Center.

Across all of these settings, I keep returning to what that physics teacher showed me: students learn best when someone makes room for their curiosity and believes they can understand. Pre-class videos, optional reels, group work at round tables, a flexible deadline, and a Zoom link on a snowy morning are all ways of acting on that belief. I will keep listening to what students tell me about their learning and adjusting my teaching in response. I hope students leave my courses with mathematics they can use, and more confident asking questions, exploring ideas, and doing mathematics with others.